

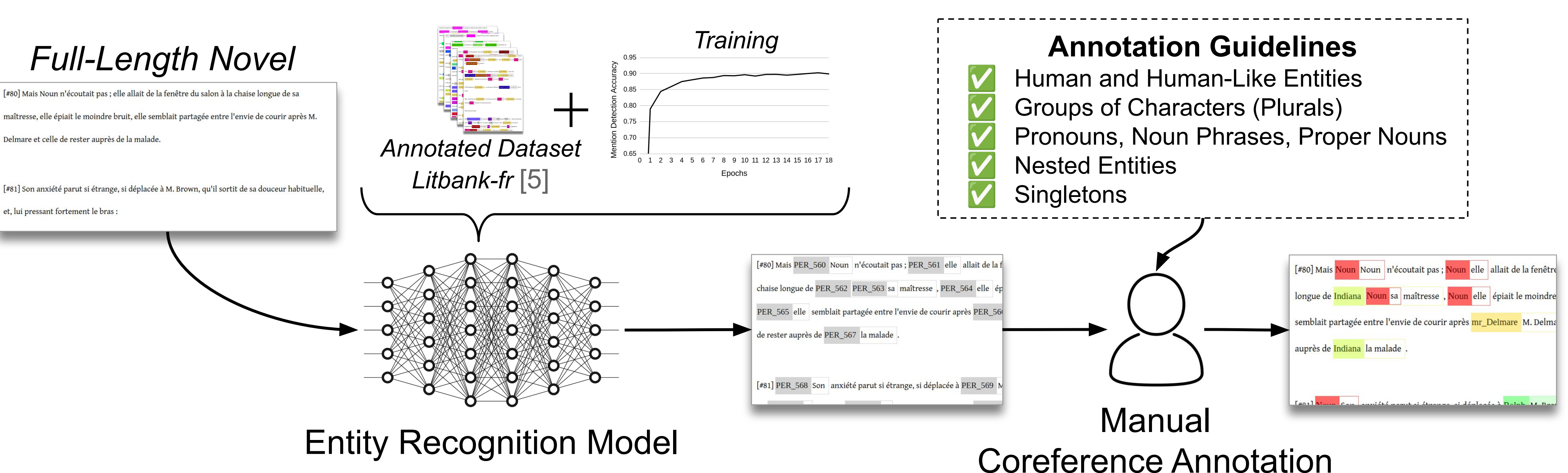
Resolving Coreference in Full-Length French Novels

- ✓ We release a dataset of 3 French novels annotated with characters coreference chains
- ✓ We study the impact of long documents on coreference resolution performance

Data Mismatch: Annotated Dataset VS Downstream Applications

Annotated Datasets	Lang.	Domain	Doc.	Tokens / Doc.	
				Average	Maximum
OntoNotes ^{en} – 2013 [1]	English	Non-literary	3,493	467	4,009
RiddleCoref – 2019 [2]	Dutch	Fiction	21	5,102	–
LitBank – 2020 [3]	English	Fiction	100	2,105	3,419
KoCoNovel – 2024 [4]	Korean	Fiction	50	3,578	19,875
LitBank-fr – 2024 [5]	French	Fiction	28	9,834	30,987
Target Datasets - Digitised Books Corpora					
Project Gutenberg	English	Fiction	61,300	110,000	1,105,964
National Library of France - Gallica	French	Fiction	15,000	100,000	878,645

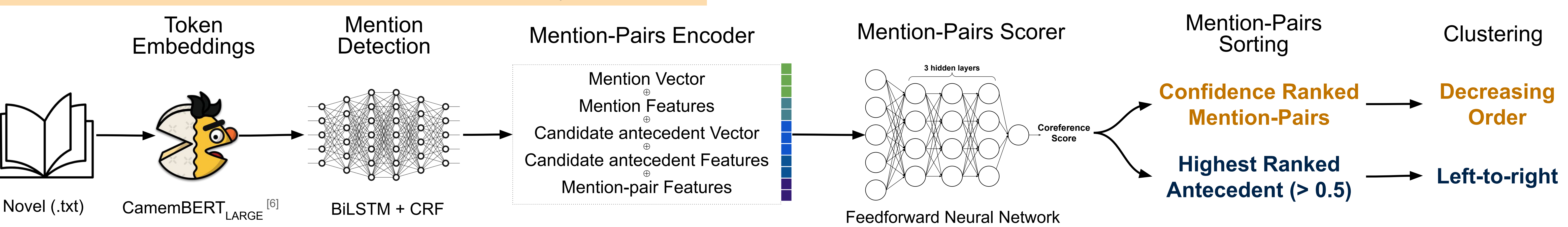
Existing annotated coreference datasets are **10× shorter** than the narratives used in real-world applications in Computational Literary Studies.



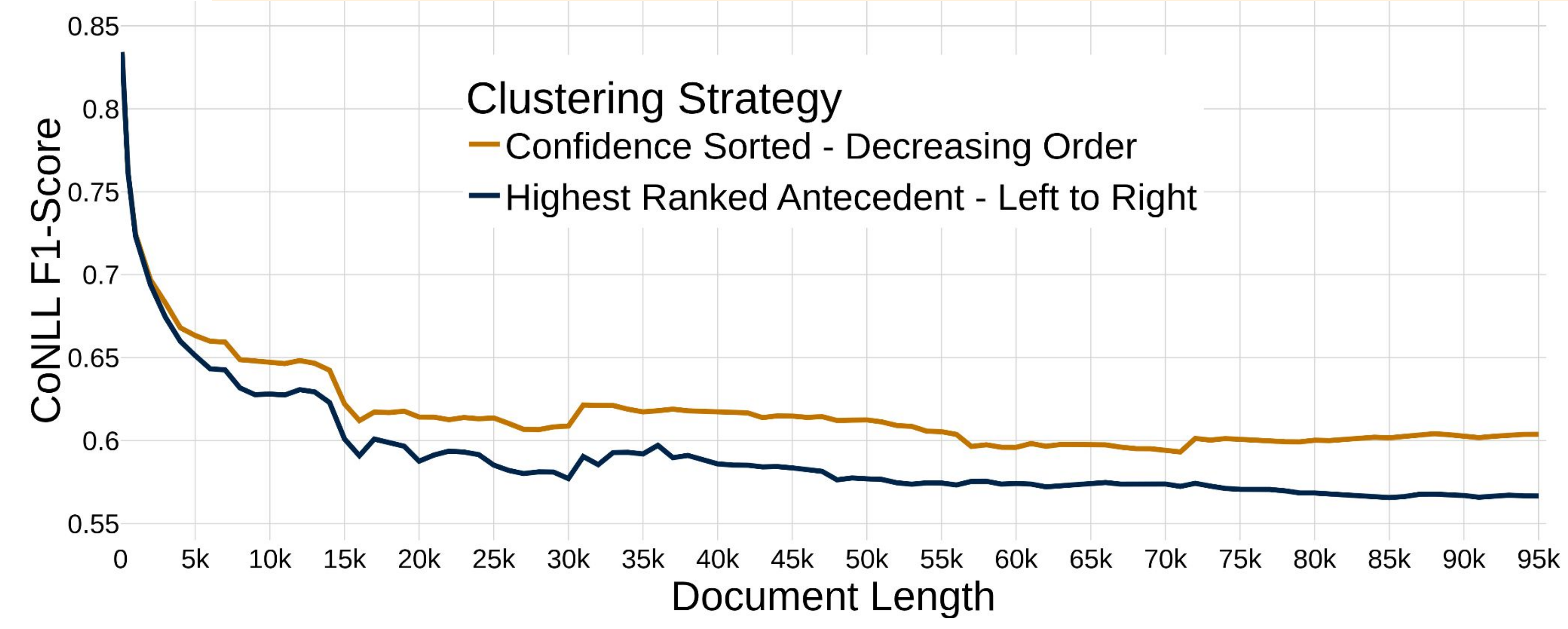
Dataset Annotation

Documents	3
Average Tokens / Document	95,058
Coreference Chains / Document	159
Average Mentions / Chain	82
Maximum Mentions / Chain	4,932
Average Entity Spread (tokens)	17,529
Maximum Entity Spread (tokens)	115,369
Pronominal Mentions	75 %

Coreference Resolution Pipeline



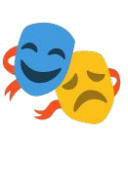
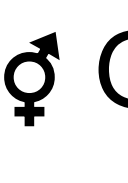
Document Length Impact on Coreference Resolution Performance



Corpus (test set)	Model	Mentions	Tokens / Doc.	CoNLL F1
LitBank-fr (test-set)	End-to-end	Gold	2,000	76.4
LitBank-fr (test-set)	Ours	Gold	2,000	79.56
Our New Dataset	Ours	Predicted	93,019	58.98
Cross-datasets Comparison				
BookCoref _{gold} [7]	Longdoc	Predicted	76,419	67.0
BookCoref _{gold} [7]	Maverick-xl	Predicted	76,419	61.0
Our New Dataset	Ours	Predicted	76,000	60.00

Coreference resolution on long narratives remains an open problem. We need new approaches and representative evaluation datasets.

Perspectives

- Multilingual Extension:  English  Italian  Russian
- Downstream Applications:  Character Archetypes,  Gender Study

Use It Now

```
pip install propp_fr

import propp_fr
propp_fr.process_text_file("your_novel.txt")
```

[1] Ralph Weischedel, Martha Palmer, Mitchell Marcus, Eduard Hovy, Sameer Pradhan, Lance Ramshaw, Nianwen Xue, Ann Taylor, Jeff Kaufman, Michelle Franchini, Mohammed El-Bachouti, Robert Belvin, and Ann Houston. 2013. OntoNotes release 5.0.
[2] Andreas van Cranenburgh. 2019. A dutch coreference resolution system with an evaluation on literary fiction. Computational Linguistics in the Netherlands Journal, 9:27–54.
[3] David Bamman, Olivia Lewke, and Anya Mansoor. 2020. An annotated dataset of coreference in English literature. In Proceedings of the Twelfth Language Resources and Evaluation Conference, pages 44–54, Marseille, France. European Language Resources Association.
[4] Kyuhee Kim, Surin Lee, and Sangah Lee. 2024. Ko-conovel: Annotated dataset of character coreference in Korean novels.
[5] Frédérique Mélanie, Jean Barré, Olga Seminck, Clément Planck, Marco Naguib, Martial Pastor, and Thierry Poibeau. 2024. Booknlp-fr, the french versant of booknlp, a tailored pipeline for 19th and 20th century french literature. Journal of Computational Literary Studies.
[6] Louis Martin, Benjamin Muller, Pedro Javier Ortiz Suárez, Yoann Dupont, Laurent Romary, Éric de la Clergerie, Djamel Seddah, and Benoît Sagot. 2020. CamemBERT: a Tasty French Language Model. Association for Computational Linguistics.
[7] Giuliano Martinelli, Tommaso Bonomo, Pere-Lluís Huguet Cabot, and Roberto Navigli. 2025. BOOKCOREF: Coreference Resolution at Book Scale. Vienna, Austria. Association for Computational Linguistics.